HOJEONG LEE

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1205 University Ave, Madison, WI 53706, USA

EDUCATION

University of Wisconsin–Madison Sep 2025 - Present
Ph.D. in Computer Sciences Madison, WI, USA
Advised by Prof. Suman Banerjee

Korea University Mar 2022 - Aug 2025
M.S. in Computer Science and Engineering Seoul, Korea
Advised by Prof. Hyogon Kim

Korea University Mar 2016 - Feb 2022
B.E. in Computer Science and Engineering Seoul, Korea

RESEARCH EXPERIENCE

WiNGS Lab, University of Wisconsin–Madison Sep 2025 - Present
Graduate Researcher Madison, WI, USA

Wireless Data Communications Lab, Korea University Mar 2022 - Aug 2025
Graduate Researcher Seoul, Korea

Internet Systems Lab, University of Colorado Boulder Mar 2024 - Mar 2025
Visiting Scholar Boulder, CO, USA
Worked with Prof. Sangtae Ha and Prof. Seyeon Kim
Fully funded by Korea University

Carnegie Mellon University Aug 2022 - Feb 2023
Visiting Student Pittsburgh, PA, USA
Selected through a merit-based national competition
Fully funded by the Korean Government

Wireless Data Communications Lab, Korea University Jun 2021 - Feb 2022
Undergraduate Research Intern Seoul, Korea

RESEARCH INTERESTS

Mobile Computing
On-device AI, energy-efficient mobile systems [W1]
Skills: GPU, NPU, speech foundation models, C++, Python

Network Systems
Volumetric video streaming [14], live video streaming, AR/VR systems, 3D gaussian splatting [W2]
Skills: C++, Python, Open3D, Draco, PCL, FFmpeg, Intel Realsense SDK

WORK IN PROGRESS

W2. **Co-author**, Online 3D Gaussian avatar refinement. Under review, 2026.

W1. **First author**, Efficient speech foundation models on mobile devices. Under review, 2026.

PUBLICATIONS

14. **Hojeong Lee**, Yu Hong Kim, Sangwoo Ryu, James Won-Ki Hong, Sangtae Ha, and Seyeon Kim. DeltaStream: 2D-Inferred Delta Encoding for Live Volumetric Video Streaming. ACM MobiSys, Anaheim, California, USA, 2025.

13. **Hojeong Lee**, Seungmo Kang, and Hyogon Kim. Causality-sensitive scheduling to reduce latency in vehicle-to-vehicle interactions. Sensors, 24 (22), 2024.

12. Seungmo Kang, **Hojeong Lee**, and Hyogon Kim. Mitigating Latency Inflation in V2C Transactions Using Periodic Sidelink Communication. IEEE VNC, Kobe, Japan, 2024.
11. **Hojeong Lee**, Chanwoo Kim, Eugene Yang, and Hyogon Kim. Distributed Joint Congestion Control for V2X Using Multi-agent Reinforcement Learning. IEEE ICMLCN, Stockholm, Sweden, 2024.
10. **Hojeong Lee** and Hyogon Kim. Improving One-Shot Transmission in NR Sidelink Resource Allocation for V2X Communication. arXiv preprint arXiv:2312.15914, 2023.
9. **Hojeong Lee** and Hyogon Kim. Rethinking Transmit Power Control for SAE J3161/1 Congestion Control Algorithm. IEEE VTC2023-Fall, Hong Kong, 2023.
8. Yeomyung Yoon, **Hojeong Lee**, and Hyogon Kim. Deep reinforcement learning-based dual-mode congestion control for cellular V2X environments. Electronics Letters, 59 (20), 2023.
7. Kyeongnam Park, **Hojeong Lee**, and Hyogon Kim. Speed-Aware V2X Congestion Control. IEEE VTC2023-Fall, Hong Kong, 2023.
6. Kyeongnam Park, Kyungha Kim, Hyungjoon Shin, **Hojeong Lee**, and Hyogon Kim. Strategically Positioning On-Board PEPs in LEO-based NTN for TCP Throughput Improvement. IEEE VTC2023-Fall, Hong Kong, 2023.
5. Hyeonji Seon, **Hojeong Lee**, and Hyogon Kim. Predicting CAM generation times through machine learning for cellular V2X communication. ICT Express, 9 (5), 2023.
4. Joseph Konan, Ojas Bhargave, Shikhar Agnihotri, **Hojeong Lee**, Ankit Shah, Shuo Han, Yunyang Zeng, Amanda Shu, Haohui Liu, Xuankai Chang, Hamza Khalid, Minseon Gwak, Kawon Lee, Minjeong Kim, and Bhiksha Raj. Improving Perceptual Quality, Intelligibility, and Acoustics on VoIP Platforms. arXiv preprint arXiv:2303.09048, 2023.
3. Hyeongji Seon, **Hojeong Lee**, and Hyogon Kim. Packet Delivery Impact of Predictive Resource Allocation for Quasi-Periodic Cellular V2X Communication. IEEE VTC2023-Spring, Florence, Italy, 2023.
2. Jonghwan Na, **Hojeong Lee**, and Hyogon Kim. Inferring Human Driver Intent in Partial Deployment of Connected Autonomous Vehicles: the Lane Change Case. IEEE VTC2023-Spring, Florence, Italy, 2023.
1. Hyeonji Seon, **Hojeong Lee**, and Hyogon Kim. Prediction of vehicle dynamics-based aperiodic message generation times in cellular V2X communication. Annual Conference of KIPS, Seoul, Korea, 2022. **OUTSTANDING PAPER AWARD**

AWARDS & GRANTS

Summer Graduate Fellowship Award, University of Wisconsin–Madison
Department of Computer Sciences, Summer 2026

Scholarships for Internationalization, Korea University
Visiting scholar funding, University of Colorado Boulder, Fall 2024

Conference Student Travel Grants
IEEE ICMLCN 2024, IEEE VTC2023-Fall

Department Merit-Based Scholarship, Korea University
Fall 2021, Spring 2021, Fall 2020

Korean Government Scholarship
Fall 2021, Spring 2021, Fall 2020, Fall 2016

TEACHING EXPERIENCE

Teaching Assistant, University of Wisconsin–Madison
Introduction to Computer Networks, Spring 2026
Foundations of Mobile Systems and Applications, Fall 2025

Teaching Assistant, Korea University
Internet Protocol, Spring 2023
Computer Network, Spring 2022

EXTRACURRICULAR ACTIVITIES

Volunteer, ACM IMC 2025 Oct 2025

Coach, NAVER Corporation Jan 2021 - Aug 2021
Boost Course PY4E (Python programming) and Harvard CS50
Selected as an excellent coach in PY4E

Software Education Volunteer, Korea University Aug 2020 - Dec 2020
Software camp instructor and mentor for middle and high school students
Software-related educational video production and Arduino project coaching

Military Service, Republic of Korea Army Jul 2018 - Mar 2020
Division Commander’s Award, 3rd Place, division combat mission-focused physical training competition

Overwatch Professional Gamer, Team LW Red Jan 2017 - Dec 2017
Runner-up, Overwatch National University Tournament Season 2 Sep 2017
Winner, Overwatch APEX Challengers Season 4 Jul 2017